SIMATS ENGINEERING
Department of Computer Science & Engineering
ASSESSMENT TOOL 2- COMPARATIVE ANALYSIS

Course Code:	**CSA12**	**Course Title:**	**Computer Architecture for Multicore Systems**
CO Assessed:	**CO4**	**Assessment:**	**ASSESSMENT TOOL2- COMPARATIVE ANALYSIS**
Weightage:	**30%**	**Total Marks:**	**25**
CO4 Statement:	*Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.*		
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Department:	**ECE**	**Date:**	**27.08.26**

ANNEXURE A
COMPARATIVE ANALYSIS

1. Compare L1, L2, and L3 cache levels in terms of latency, bandwidth, and throughput, and analyze their impact on processor performance.
2. Differentiate SRAM and DRAM by evaluating their latency, bandwidth, and throughput characteristics in cache and main memory design.
3. Compare virtual memory with paging and cache memory mechanisms in terms of access latency and system throughput.
4. Analyze the differences between NAND Flash and DRAM with respect to latency, bandwidth, and suitability for hierarchical memory systems.
5. Compare MRAM and RRAM technologies by examining their performance in terms of access time, throughput, and energy efficiency.

Code of Conduct:

I GOPIKA P (192412306)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:GOPIKA P

MARKS ALLOCATION

	Scope of Items (20%)	Comparison Depth (25%)	Use of Evidence (20%)	Critical Thinking (20%)	Clarity of Presentation (15%)
Q1					
Q2					
Q3					
Q4					
Q5					

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Answer:

- Latency: L1 (~1–2 cycles), L2 (~3–10 cycles), L3 (~30–50 cycles).
- Bandwidth: L1 highest, L2 high, L3 moderate.
- Throughput: L1 ensures fastest reuse, L2 balances speed/size, L3 reduces main memory calls.
- Impact: Larger caches reduce DRAM access, improving processor performance.

Answer:

- Latency: L1 extremely low, L2 moderate, L3 higher.
- Bandwidth: L1 peak bandwidth, L2 strong, L3 shared across cores.
- Throughput: L1 critical for instruction/data reuse, L2 for mid-level reuse, L3 for inter-core communication.
- Impact: Cache hierarchy minimizes stalls, but L3 latency can bottleneck multi-core workloads.

Answer:

- Latency: L1 fastest, L2 slower, L3 slowest but still faster than DRAM.
- Bandwidth: L1 per-core maximum, L2 good balance, L3 shared bandwidth.
- Throughput: L1 drives single-thread speed, L2 supports moderate workloads, L3 improves multi-core throughput.
- Impact: Hierarchical caches optimize both single-core and multi-core performance.

CONCLUSION: gave the better comparison because it included exact latency cycle values along with bandwidth and throughput, making the answer more precise. Claude and Gemini explained well, but numerical detail makes its answer clearer and stronger.

2. SRAM vs DRAM

Answer:

- SRAM: Latency ~1 ns, very high bandwidth, high throughput, volatile. Used in caches.
- DRAM: Latency ~50–100 ns, high bandwidth, lower throughput, volatile. Used in main memory.
- Design Role: SRAM for speed-critical cache, DRAM for bulk storage.

Answer:

- SRAM: Faster, consumes more power, expensive, ideal for small high-speed caches.
- DRAM: Slower, cheaper, denser, suitable for large main memory.
- Design Role: SRAM ensures low-latency cache hits; DRAM balances cost and capacity.

Answer:

- SRAM: Ultra-low latency, limited density, high cost. Best for CPU caches.
- DRAM: Higher latency, scalable density, cost-effective. Best for main memory.
- Design Role: SRAM accelerates computation; DRAM provides capacity.

CONCLUSION: gave the better comparison for SRAM vs DRAM because it clearly showed latency, density, and cost trade-offs while linking them to usage — SRAM for speed and DRAM for capacity. added numbers and Claude noted cost/power, but Gemini's balance makes its answer stronger.

3. Virtual Memory vs Cache

Answer:

- Virtual Memory (Paging): Latency in microseconds (disk/SSD), throughput limited.
- Cache Memory: Latency nanoseconds, throughput high.
- Comparison: Cache accelerates CPU operations; paging extends memory capacity but slows performance.

Answer:

- Virtual Memory: Provides abstraction, but paging incurs high latency and reduces throughput.
- Cache Memory: Hardware-managed, extremely low latency, boosts throughput.
- Comparison: Cache improves speed; virtual memory improves capacity and isolation.

Answer:

- Virtual Memory: Extends address space, but paging overhead reduces throughput.
- Cache Memory: Directly reduces CPU stalls, nanosecond latency.
- Comparison: Cache is performance-oriented; virtual memory is capacity-oriented.

CONCLUSION: gave the better comparison because it included specific latency values in microseconds vs nanoseconds and tied them directly to throughput, making the answer more precise. Claude and Gemini explained the concepts clearly, but numerical detail makes its comparison stronger and more exam-ready.

4. NAND Flash vs DRAM

Answer:

- NAND Flash: Latency microseconds, bandwidth moderate, non-volatile. Best for storage.
- DRAM: Latency ~100 ns, bandwidth very high, volatile. Best for working memory.
- Suitability: NAND for persistence, DRAM for execution.

Answer:

- NAND Flash: Non-volatile, slower, cheaper per bit, ideal for long-term storage.
- DRAM: Volatile, faster, higher bandwidth, ideal for active workloads.
- Suitability: NAND complements DRAM in hierarchical systems.

Answer:

- NAND Flash: Non-volatile, slower access, high density.
- DRAM: Volatile, fast, lower density.
- Suitability: NAND for storage tiers, DRAM for execution tiers.

CONCLUSION: gave the better comparison here because it not only explained the speed and volatility differences but also added cost and system design context, showing how NAND and DRAM complement each other in hierarchical memory. gave useful latency values and Gemini highlighted density, but Claude's balance of technical and practical detail makes its answer stronger.

5. MRAM vs RRAM

Answer:

- MRAM: Latency ~10 ns, throughput moderate, energy efficiency good, non-volatile.
- RRAM: Latency ~10 ns, throughput high, energy efficiency very high, non-volatile.
- Comparison: MRAM reliable for persistence, RRAM promising for ultra-low-power inference.

Answer:

- MRAM: Magnetic storage, fast, durable, moderate energy efficiency.
- RRAM: Resistive switching, faster writes, higher density, excellent energy efficiency.
- Comparison: MRAM stable and mature; RRAM experimental but potentially superior in efficiency.

Answer:

- MRAM: Non-volatile, fast reads, moderate density, good endurance.
- RRAM: Non-volatile, scalable density, excellent energy efficiency, emerging tech.
- Comparison: MRAM is practical today; RRAM is future-oriented.

CONCLUSION: gave the better comparison for MRAM vs RRAM because it explained maturity, durability, and efficiency trade-offs clearly. added latency values and Gemini gave future perspective, but balance of technical detail and practical context makes its answer stronger.